Design and Implementation of a 2-Bit Digital Counter Using a 555 Timer and Logic Gates

College Hackathon Project Documentation

Project type	Hardware-based digital electronics
Main concepts	555 timer, NAND/NOT gates, flip-flop logic, 2-bit counter
Display	Common-cathode 7-segment display
Observed output	0 → 1 → 2 → 3 → repeat

Prepared from the project description provided. Add your name, college, department, event name/date, and actual circuit photographs before submission.

Abstract

This project demonstrates a hardware implementation of a 2-bit digital counter. A 555 timer IC provides clock pulses to a sequential logic circuit built using NAND and NOT gates to implement flip-flop functionality. The two-bit state is decoded through logic gates to drive a common-cathode 7-segment display, displaying the decimal sequence 0, 1, 2 and 3. The reported hardware demonstration cycles through these four digits. The exact 555 timer operating mode, component values, supply voltage, and measured pulse frequency were not recorded and are therefore not specified in this report.

1. Aim

To build a hardware 2-bit digital counter using a 555 timer, flip-flop logic implemented with NAND and NOT gates, and a common-cathode 7-segment display to show the sequence 0–3.

2. Objectives

• Learn the role of an integrated circuit (IC) and a 7-segment display.
• Generate clock pulses using a 555 timer IC.
• Implement the required state-storage logic using NAND and NOT gates.
• Produce and observe the four states of a 2-bit binary counter.
• Decode the binary states to display decimal digits on a common-cathode 7-segment display.
• Gain practical experience in hardware assembly, troubleshooting, and learning unfamiliar concepts under time constraints.

3. Hardware Components

The project description identifies the following items. Verify the exact part numbers and quantities against the circuit before final submission:
• 555 timer IC
• NAND gates and NOT gates (or corresponding logic ICs)
• Components and connections required to form the flip-flop stages
• Common-cathode 7-segment display
• Resistors and capacitor(s) used by the timer circuit
• Breadboard or other circuit assembly board
• Jumper/connecting wires
• Suitable DC power supply
• Current-limiting resistors and any additional interface components used in the actual circuit

4. Block-Level Operation

555 timer clock pulses → flip-flop/state logic → 2-bit binary outputs → decoding logic gates → common-cathode 7-segment display.

5. Working Principle

5.1 Clock pulse generation: The 555 timer provides pulses that advance the counter. A 555 timer can operate in astable or monostable mode; the mode used in this build has not been confirmed. Timing frequency depends on the actual circuit and component values, which are not specified here.

5.2 State storage: NAND and NOT gates are used to implement the flip-flop functionality described for the project. The exact gate-level topology should be documented using the final circuit diagram.

5.3 Two-bit counting: Two binary output bits can represent four states: 00, 01, 10 and 11. The observed display sequence is 0, 1, 2 and 3, then repeats. The table below shows the intended state-to-digit mapping;

confirm bit labels against the actual wiring.

5.4 Display decoding: Combinational logic converts the counter outputs into segment-control signals. For a common-cathode display, the common cathode is connected to the appropriate ground reference and segments generally illuminate when driven HIGH through suitable current-limiting/interface circuitry. Confirm the actual pinout and resistor arrangement before powering the circuit.

6. State Table

Count step	Binary output (Q1 Q0)	Displayed digit
1	00	0
2	01	1
3	10	2
4	11	3
Next cycle	00	0

The bit labels are illustrative; align Q1 and Q0 with the actual wiring in the final schematic.

7. Result

The physical hardware demonstration was reported to cycle through the digits 0, 1, 2 and 3 on a common-cathode 7-segment display. This is consistent with the four possible states of a 2-bit counter. No measured clock frequency, timing-component values, or formal test measurements were provided.

8. Learning Outcomes

This hackathon provided an opportunity to learn basic electronics concepts from online resources and with guidance from college seniors, then apply those concepts in a hardware task. The work introduced practical ideas related to ICs, 555 timer pulse generation, logic gates, flip-flop state storage, binary counting, display interfacing, and circuit troubleshooting.

9. Conclusion

The project demonstrates a hardware approach to a 2-bit counter using a 555 timer, NAND/NOT-based flip-flop logic, decoding gates, and a common-cathode 7-segment display. The observed 0–3 sequence connects binary states with decimal display output. The experience strengthened practical learning and problem-solving skills. The exact timer mode, component values, and gate-level implementation should be confirmed from the original circuit before this report is treated as a complete circuit specification.

10. Possible Future Improvements

- Add a labelled final schematic with gate names, pin numbers, and signal labels.
- Record the timer configuration, resistor/capacitor values, supply voltage, and measured frequency.
- Add a reset input if it is not already present.
- Test and document each binary state and its display output.
- Extend the counter with additional stages and suitable decoding logic.

Acknowledgements

I would like to thank my college seniors for helping me understand the basic concepts and my professors for their encouragement and appreciation during the college hackathon.

References

Texas Instruments, NE555 product page and datasheet: https://www.ti.com/product/NE555
Use the datasheet and the actual circuit diagram to verify pin connections and timing details. No unverified component values are assumed in this report.